

EXPERIENCE

BeyPark, Beypore International Water Fest

Beypore, India

*Software Developer**December 2025*

- Designed and implemented an autonomous parking management system to handle high traffic for a major international water test event.
- Architected a master-slave distributed system linking multiple edge cameras and remote parking lots to a centralized database for real-time capacity monitoring.

Oracle

Hyderabad, India

*Application Developer Summer Intern**May 2025 – July 2025*

- Collaborated with enterprise teams to develop and optimize scalable software solutions.
- Engineered backend microservices and improved API response times through efficient data handling.
- Participated in agile code reviews and system design sessions to ensure high-quality production deployments.

Computer Science and Engineering Department (CSED), NITC

Calicut, India

*Team Lead, Website Redevelopment**November 2024 – January 2025*

- Directed a technical team in the complete overhaul of the departmental website using a modern technology stack.
- Migrated the frontend architecture to Next.js, significantly improving page load speeds and SEO metrics.
- Rebuilt the backend utilizing Strapi as a headless CMS to streamline content management for faculty and administration.

PROJECTS

Sentinel (ANPR & Traffic System) | Python, YOLO, OCR, Microservices*2025 – 2026*

- Engineered a microservices-based Automatic Number Plate Recognition (ANPR) and traffic violation detection system.
- Integrated YOLO and various OCR models for real-time edge monitoring, drafting a comprehensive deployment proposal for 9 active locations across Calicut.
- Co-authored and presented a research paper detailing the system's core OCR equations and optimization heuristics at an institute conference.

HQC Algorithm Optimization | C++, AVX2, SIMD Vectorization*2025 – 2026*

- Co-authored cryptographic optimization research (yet to be published) focusing on the Hamming Quasi-Cyclic (HQC) algorithm for x86 architecture.
- Leveraged AVX2 instructions and memory prefetching to optimize computational bottlenecks in core operations.
- Achieved an 11.02% reduction in decapsulation cycles for HQC-128 by implementing a highly efficient LUT-based field inversion.

NITCBase | C++*2025*

- Developed a lightweight relational database management system from scratch.
- Implemented core functionalities including dynamic schema definition, sophisticated query parsing, and persistent data storage.

UniVerse | Node.js, PostgreSQL, React, Tailwind CSS*May 2024*

- Built a secure, full-stack student forum platform to facilitate campus communication and resource sharing.
- Integrated JWT-based authentication and automated email verification via Nodemailer to ensure secure user onboarding.

Automated X/Twitter Bot | JavaScript, API Integration*2025*

- Developed and deployed a scheduling automation script that dynamically posts programming content every 2 hours.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL**Frameworks & Tools:** React, Next.js, Node.js, Java, Bash, Git, Oracle Visual Builder**Spoken Languages:** English, Marathi, Hindi

EDUCATION

National Institute of Technology Calicut (NITC)

Calicut, India

*B.Tech in Computer Science and Engineering; CGPA: 7.99**2022 – 2026***D.K.T.E. Jr. College**

Ichalkaranji, India

*HSC (Science); 86.33%**2020 – 2022*

EXTRACURRICULARS & LEADERSHIP

System Design Workshop Instructor

NITC

*Technical Speaker**January 2026 – February 2026*

- Conducted comprehensive workshops for peers utilizing process flowcharts to explain complex software architectures.
- Leveraged Sentinel and BeyPark as real-world case studies to teach practical microservices and edge-computing logic.

Computer Science and Engineering Association (CSEA)

NITC

*Treasurer**2025 – 2026*

- Managed the organization's financial operations, overseeing budgeting and advance payments for large-scale technical events.

Ekdant 25, Ganeshotsav Celebration

NITC

*Executive Member**2024*

- Coordinated cross-functional logistics and event planning for a major cultural festival handling a footfall of 2000+ attendees.